



Highway to Renewable Energy Systems

H2RES v1.0 Report (Draft)

Mathematical Formulation and Data Dictionary

Document Type: Technical Document

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1 Scope

This document reconstructs the optimization model implemented in `scripts/Build_model.py` and the data processing logic in `scripts/read_process_data.py`. It lists all decision variables and all symbols used as parameters/sets/auxiliary terms, then explains the objective and each constraint block.

2 Model Architecture Diagram

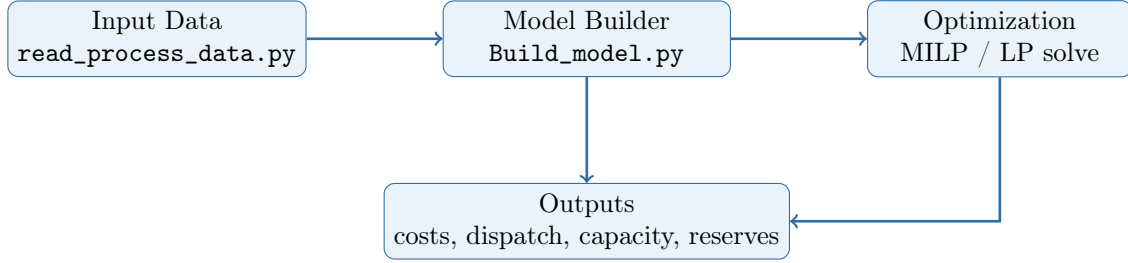


Figure 1: High-level workflow of the H2RES optimization model.

3 H2RES Model

The H2RES model is an energy system modeling tool developed within the framework of the INTERENERGY project at the University of Zagreb. Its main objective is to analyze energy transition pathways by representing and optimizing different components of the energy system. The model is particularly oriented towards the assessment of flexibility technologies, sector coupling, demand response, and power to X solutions, including the production of hydrogen.

H2RES allows the assessment of both energy system operation and its evolution over time. Compared with models primarily focused on the simulation of a predefined energy system configuration, H2RES allows the analysis of investments in new capacities, the gradual decommissioning of existing technologies, hourly system operation, and technical or economic constraints. This makes it a useful tool for studying long term scenarios associated with decarbonisation and the integration of variable renewable energy sources.

The model represents several sectors of the energy system, including electricity, transport, industry, heating and cooling, hydrogen, energy storage, and CO₂ emissions. In the electricity sector, it can simulate conventional and renewable generation, imports, exports, storage, excess renewable electricity, and operational constraints.

A central feature of H2RES is its representation of storage and flexibility technologies. The model includes stationary batteries, pumped hydropower storage, thermal energy storage, electric vehicle batteries, hydrogen storage, and technologies associated with the conversion of electricity into other energy carriers. This capability is particularly relevant for analysing systems with high shares of solar and wind energy, where flexibility is required to balance the variability of renewable generation with energy demand.

Overall, H2RES can be understood as an hourly resolution energy planning model designed to evaluate transition pathways towards more renewable, flexible, and low emission energy systems. Its open source nature allows users to review, adapt, and improve its equations and structure, thereby supporting methodological transparency and application across different research contexts. For these

reasons, H2RES is a relevant tool for the academic analysis of energy scenarios, sector coupling, electrification, hydrogen, and power to X strategies.

Figure 2 presents the general schematic structure of the H2RES model, illustrating how different energy sectors and technologies are interconnected within a single modeling framework. The figure highlights the interaction between electricity generation, demand sectors, storage technologies, hydrogen production, transport, industry, and heating and cooling systems. It also shows the role of power to X pathways and flexibility options in linking renewable electricity supply with different final energy demands. Overall, the figure provides a conceptual overview of H2RES as an integrated energy system model designed to analyse sector coupling, storage, and energy transition pathways.

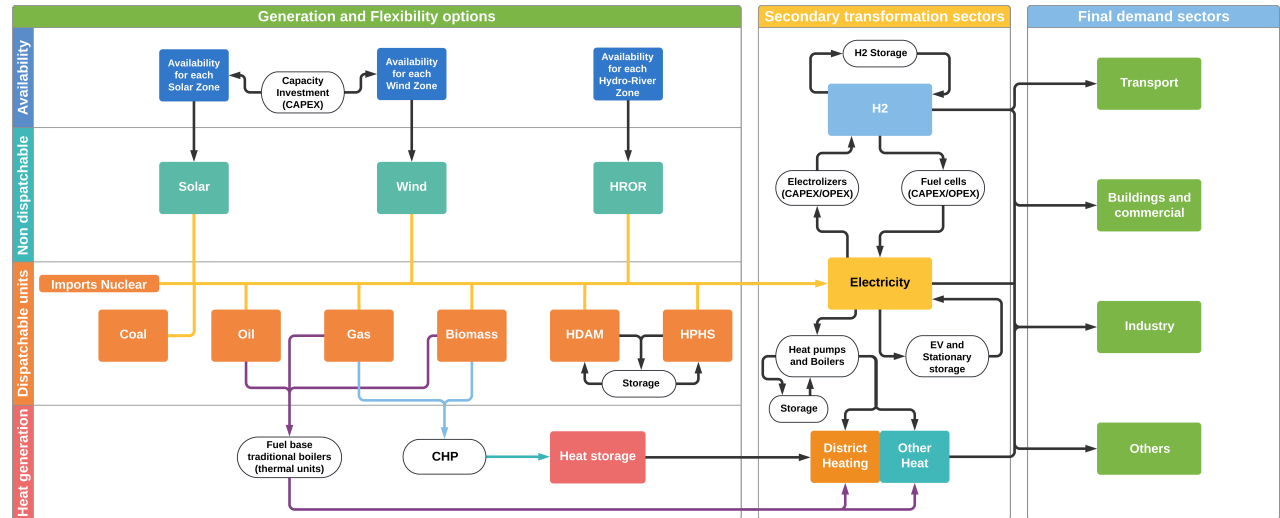


Figure 2: H2RES model schematics

4 Data Dictionary Tables (Complete)

4.1 Decision Variables (Complete)

Variable (original model notation)	Type	Unit	Description
$\text{rampup_cost}_{disp_units,t,y}$	Cost auxiliary	EUR	Auxiliary upward ramping cost variable for dispatchable units.
$\text{rampdown_cost}_{disp_units,t,y}$	Cost auxiliary	EUR	Auxiliary downward ramping cost variable for dispatchable units.
$\text{generation}_{u,t,y}$	Operation	MW _e (hourly average)	Hourly electric generation by unit.
$\text{imports}_{t,y}$	Operation	MW _e (hourly average)	Hourly electricity imports.
$\text{ceep}_{t,y}$	Operation	MW _e (hourly average)	Curtailement / non-served energy slack variable (ceep).
$\text{cap_inv}_{u,y}$	Investment	MW _e	Annual generation capacity investment.
CO2_total_y	Emissions	tCO ₂ /year	Annual CO ₂ emissions from electric generation.
CO2_total_heat_y	Emissions	tCO ₂ /year	Annual CO ₂ emissions from heat supply.
$\text{CO2_total_industry}_y$	Emissions	tCO ₂ /year	Annual CO ₂ emissions from industrial fuels.
$\text{cap_inv_FC}_{FC_units,y}$	Investment	MW _e	Annual fuel-cell capacity investment.
$\text{generation_FC}_{FC_units,t,y}$	Operation	MW _e (hourly average)	Hourly electric generation from fuel cells.
$\text{storage_level}_{hydro_storage_units,t,y}$	State variable	MW _h	Hydro storage level (HDAM/HPHS).
$\text{sto_out_flow}_{hydro_storage_units,t,y}$	Operation	MW _e eq.	Hydro spill / outflow not converted to electricity.
$\text{heat_gen}_{chp_units,t,y}$	Operation	MW _{th}	Heat output from CHP units.
$\text{heat_input}_{chp_units,t,y}$	Operation	MW _{th} eq.	Electric-to-heat extraction input in CHP coupling.
$\text{heat_sto_level}_{chp_units,t,y}$	Operation	MW _h _{th}	CHP thermal storage state of charge.
$\text{Fuel_ind}_{industry_fuels,t,y}$	Operation	MW _{th}	Industrial fuel use by fuel type.
$\text{H2tInd}_{t,y}$	Operation	MW _{H2}	Hydrogen use in industrial high-temperature demand.
$\text{heat_pump_out}_{HeatPump_units,t,y}$	Operation	MW _{th}	Heat-pump operation or expansion variable by market/segment.
$\text{heat_pump_dh}_{HeatPump_units,chp_units,t,y}$	Operation	MW _{th}	Heat-pump operation or expansion variable by market/segment.
$\text{heat_pump_ind}_{HeatPump_units,t,y}$	Operation	MW _{th}	Heat-pump operation or expansion variable by market/segment.
$\text{heat_pump_dh_cap}_{HeatPump_units,chp_units,y}$	Investment	MW _{th}	Heat-pump operation or expansion variable by market/segment.
$\text{heat_pump_ind_cap}_{HeatPump_units,y}$	Investment	MW _{th}	Heat-pump operation or expansion variable by market/segment.
$\text{cooling_pump_dh}_{HeatPump_units,chp_units,t,y}$	Operation	MW _{cool}	Cooling output from heat-pump technologies.
$\text{cooling_pump_ind}_{HeatPump_units,t,y}$	Operation	MW _{cool}	Cooling output from heat-pump technologies.
$\text{boiler_dh_generation}_{thermal_units,chp_units,t,y}$	Operation	MW _{th}	Thermal boiler heat production variable.
$\text{boiler_ind_generation}_{thermal_units,t,y}$	Operation	MW _{th}	Thermal boiler heat production variable.
$\text{trad_boiler_ind_cap_inv}_{thermal_units,y}$	Investment	MW _{th}	Thermal boiler capacity investment variable.
$\text{trad_boiler_dh_cap_inv}_{thermal_units,chp_units,y}$	Investment	MW _{th}	Thermal boiler capacity investment variable.
$\text{ResToHeat}_{t,y}$	Operation	MW _e (hourly average)	Electricity converted to heat/cooling services (power-to-heat).
$\text{heat_pump_cap}_{HeatPump_units,y}$	Investment	MW _{th}	Heat-pump operation or expansion variable by market/segment.
$\text{heat_pump_sto_soc}_{HeatPump_units,t,y}$	State variable	MW _h _{th}	Heat-pump operation or expansion variable by market/segment.
$\text{ev_sto_in}_{t,y}$	Operation	MW _e	EV fleet charging/discharging/storage operation variable.

Variable (original model notation)	Type	Unit	Description
ev_sto_out _{t,y}	Operation	MW _e	EV fleet charging/discharging/storage operation variable.
ev_sto_soc _{t,y}	State variable	MWh _e	Aggregate EV fleet state of charge.
stat_sto_in _{Stat_storage_unit,t,y}	Operation	MW _e	Stationary battery operation/investment variable.
stat_sto_out _{Stat_storage_unit,t,y}	Operation	MW _e	Stationary battery operation/investment variable.
stat_sto_soc _{Stat_storage_unit,t,y}	State variable	MWh _e	Stationary battery state of charge.
stat_sto_cap_inv _{Stat_storage_unit,y}	Investment	MWh _e	Stationary battery operation/investment variable.
power_to_h2 _{t,y}	Operation	MW _e	Total electric load used for hydrogen production.
h2_sto_soc _{H2_storage_unit,t,y}	State variable	MWh _{H2}	Hydrogen storage state of charge.
h2_sto_out _{H2_storage_unit,t,y}	Operation	MW _{H2}	Hydrogen storage operation/investment variable.
h2_sto_max_cap _{H2_storage_unit,y}	Investment	MWh _{H2}	Hydrogen storage operation/investment variable.
h2_electrolysis_max_cap _{Elec_h2_units,y}	Investment	MW _{H2}	Decision variable used in the optimization model.
generation_H2 _{Elec_h2_units,t,y}	Operation	MW _{H2}	Hydrogen production by electrolyzers.
tot_electric_load _{t,y}	Auxiliary	MW _e	Decision variable used in the optimization model.
max_cap_installed _y	Auxiliary	MW _e	Decision variable used in the optimization model.

4.2 Parameters, Sets, and Auxiliary Terms (Complete)

Parameter / set (original model notation)	Category	Unit	Description
Boiler_decom_rate _{thermal_units}	Technical	p.u. per hour/year	Capacity decommissioning/degradation rate.
Boiler_decom_start_new _{thermal_units}	Technical	year	Thermal boiler module parameter.
CHPMaxHeat _{chp_units}	Technical	MW	CHP coupling / thermal storage parameter.
CHPPowerLossFactor _{chp_units}	Technical	p.u.	CHP coupling / thermal storage parameter.
CHPPowerToHeat _{chp_units}	Technical	p.u.	CHP coupling / thermal storage parameter.
CO2_cost_industry	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
CO2_emissions_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
CO2_factor _u	Technical	tCO2/MWh	CO2 intensity factor.
CO2_limit _y	Policy	tCO2/year	Policy parameter by model year.
CO2_price _y	Policy	EUR/tCO2	Policy parameter by model year.
Elec_h2_units	Set	-	Index set/subset used in constraints and objective terms.
FC_eff _{FC_units}	Technical	p.u.	Hydrogen and fuel-cell module parameter.
FC_init_cap _{FC_units}	Technical	MW	Hydrogen and fuel-cell module parameter.
FC_max_cap _{FC_units}	Technical	MW	Hydrogen and fuel-cell module parameter.
FC_units	Set	-	Index set/subset used in constraints and objective terms.
FC_var_cost _{FC_units}	Technical	EUR/MWh	Hydrogen and fuel-cell module parameter.
Fossil_hydro	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
Fuel_CO2 _{industry_fuels}	Technical	tCO2/MWh	CO2 intensity factor.
H2_demand _{demand_sectors,t,y}	Demand	MW _{H2}	Exogenous demand trajectory by market/sector and time.

Parameter / set (original model notation)	Category	Unit	Description
H2_eff _{Elec_h2_units}	Technical	p.u.	Hydrogen and fuel-cell module parameter.
H2_init_cap _{Elec_h2_units}	Technical	MW	Hydrogen and fuel-cell module parameter.
H2_max_cap _{Elec_h2_units}	Technical	MW	Hydrogen and fuel-cell module parameter.
H2_sto_init _{H2_storage_unit}	Technical	MWh	Hydrogen and fuel-cell module parameter.
H2_sto_max_cap _{H2_storage_unit}	Technical	MWh	Hydrogen and fuel-cell module parameter.
H2_storage_unit	Technical	model units	Hydrogen and fuel-cell module parameter.
H2_var_cost _{Elec_h2_units}	Technical	EUR/MWh	Hydrogen and fuel-cell module parameter.
HP_Boiler_dh_init_cap _{HeatPump_units,chp_units}	Technical	MW	Heat pump / heat system parameter.
HP_COP_period _{HeatPump_units,t,y}	Technical	p.u.	Heat pump / heat system parameter.
HP_Decom_ind _{HeatPump_units,y}	Technical	p.u.	Heat pump / heat system parameter.
HP_decom_rate _{HeatPump_units}	Technical	p.u. per hour/year	Capacity decommissioning/degradation rate.
HP_decom_start_new _{HeatPump_units}	Technical	year	Heat pump / heat system parameter.
HP_ind_init_cap _{HeatPump_units}	Technical	MW	Heat pump / heat system parameter.
HP_max_cap _{HeatPump_units}	Technical	MW	Heat pump / heat system parameter.
HP_sto_init _{HeatPump_units}	Technical	MWh	Heat pump / heat system parameter.
HP_var_cost _{HeatPump_units}	Technical	model units	Heat pump / heat system parameter.
HeatPump_units	Set	-	Index set/subset used in constraints and objective terms.
Imp_price_inc _y	Economic	% per year	Exogenous economic parameter.
Import_Price _{t,y}	Economic	EUR/MWh _e	Exogenous economic parameter.
Industry_HT_demand _{t,y}	Demand	MW-equivalent (hourly)	Exogenous demand trajectory by market/sector and time.
NPV	Economic	p.u.	Exogenous economic parameter.
NoResToHeatInv	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
ResToHeat_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
STOCapacity_heat _{chp_units}	Technical	MWh	CHP coupling / thermal storage parameter.
StaSto_Decom_ind _y	Technical	p.u.	Exogenous parameter or helper alias used in constraints/objective.
Sta_sto_decom_rate _{Stat_storage_unit}	Technical	p.u. per hour/year	Capacity decommissioning/degradation rate.
Sta_sto_decom_start_new _{Stat_storage_unit}	Technical	year	Stationary storage module parameter.
Sta_sto_eff _{Stat_storage_unit}	Technical	p.u.	Stationary storage module parameter.
Sta_sto_init_cap _{Stat_storage_unit}	Technical	MWh	Stationary storage module parameter.
Sta_sto_max_cap _{Stat_storage_unit}	Technical	MWh	Stationary storage module parameter.
Sta_sto_var_cost _{Stat_storage_unit}	Technical	model units	Stationary storage module parameter.
Stat_storage_unit	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
TechChange	Technical	p.u.	Capacity expansion/decommissioning trajectory parameter.
Thermal_CO2 _{thermal_units}	Technical	tCO2/MWh	CO2 intensity factor.
Thermal_Decom_DH _{thermal_units,chp_units,y}	Technical	p.u.	Thermal boiler module parameter.
Thermal_Decom_ind _{thermal_units,y}	Technical	p.u.	Thermal boiler module parameter.
Thermal_ind_init_cap _{thermal_units}	Technical	MW	Thermal boiler module parameter.
Thermal_max_cap _{thermal_units}	Technical	MW	Thermal boiler module parameter.
V2G_cost _{t,y}	Economic	EUR/MWh _e	Exogenous economic parameter.
aval_factor_plant _{u,t,y}	Technical	p.u.	Hydro inflow/efficiency/availability parameter.
biomass_units	Set	-	Index set/subset used in constraints and objective terms.

Parameter / set (original model notation)	Category	Unit	Description
boiler_gen_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
cap_factor _u	Technical	p.u.	Technical/economic parameter for generation technologies.
cap_inv_cost _{u,y}	Technical	EUR/MW	Technical/economic parameter for generation technologies.
carbonLimit	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
ceep_demand _y	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
ceep_limit	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
ceep_parameter _y	Policy	p.u.	Upper bound on annual ceep as share of annual demand.
chp_units	Set	-	Index set/subset used in constraints and objective terms.
cooling_demand _{demand_sectors,t,y}	Demand	MW-equivalent (hourly)	Exogenous demand trajectory by market/sector and time.
decom_rate _u	Technical	p.u. per hour/year	Capacity decommissioning/degradation rate.
decom_start_new _u	Technical	year	Capacity expansion/decommissioning trajectory parameter.
decom_start_old _u	Technical	year	Capacity expansion/decommissioning trajectory parameter.
demand _{demand_sectors,t,y}	Demand	MW-equivalent (hourly)	Exogenous demand trajectory by market/sector and time.
demand_sectors	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
disp_units	Set	-	Index set/subset used in constraints and objective terms.
evToPower _t _cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
ev_Demand _y	Technical	MW-equivalent (hourly)	EV/V2G module parameter.
ev_Grid_P_max _y	Technical	MW	EV/V2G module parameter.
ev_Grid_eff _y	Technical	p.u.	EV/V2G module parameter.
ev_availability _{t,y}	Technical	p.u.	EV/V2G module parameter.
ev_sto_max _y	Technical	MWh	EV/V2G module parameter.
ev_sto_min _y	Technical	p.u.	EV/V2G module parameter.
ev_transp_load _{t,y}	Technical	MW-equivalent (hourly)	EV/V2G module parameter.
exports _{t,y}	Technical	MW _e	Interconnection/import-export parameter by period.
exports_dat	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
fossil_units	Set	-	Index set/subset used in constraints and objective terms.
fuel_share_max _{t,y,industry_fuels}	Technical	p.u.	Exogenous parameter or helper alias used in constraints/objective.
h2_CAPEX_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
h2_OPEX_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
h2_storage_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
hdam_efficiency _u	Technical	p.u.	Hydro inflow/efficiency/availability parameter.
hdam_inflow _{u,t,y}	Technical	inflow p.u.	Hydro inflow/efficiency/availability parameter.
hdam_max_storage _u	Technical	MWh	Hydro inflow/efficiency/availability parameter.
hdam_units	Set	-	Index set/subset used in constraints and objective terms.
heat_demand _{chp_units,t,y}	Demand	MW-equivalent (hourly)	Exogenous demand trajectory by market/sector and time.
heat_pump_cap_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.

Parameter / set (original model notation)	Category	Unit	Description
heat_var_cost _{HeatPump_units}	Technical	EUR/MWh	Heat pump / heat system parameter.
hphs_efficiency _u	Technical	p.u.	Hydro inflow/efficiency/availability parameter.
hphs_inflow _{u,t,y}	Technical	inflow p.u.	Hydro inflow/efficiency/availability parameter.
hphs_max_storage _u	Technical	MWh	Hydro inflow/efficiency/availability parameter.
hphs_units	Set	-	Index set/subset used in constraints and objective terms.
hror_units	Set	-	Index set/subset used in constraints and objective terms.
hydro_storage	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
hydro_storage_units	Set	-	Index set/subset used in constraints and objective terms.
hydro_units	Set	-	Index set/subset used in constraints and objective terms.
import_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
import_ntc _t	Technical	MW _e	Interconnection/import-export parameter by period.
industry_fuel_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
industry_fuels	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
industry_fuels_price _{industry_fuels,t,y}	Technical	EUR/MWh _{fuel}	Exogenous parameter or helper alias used in constraints/objective.
inv_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
inv_cost_FuelCell	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
inv_cost_boiler	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
inv_cost_fossil	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
max_inv_period _u	Technical	MW	Capacity expansion/decommissioning trajectory parameter.
max_load _u	Technical	MW _e	Technical/economic parameter for generation technologies.
nondisp_units	Set	-	Index set/subset used in constraints and objective terms.
per_mw	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
per_mw_FC	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
periods	Set	-	Index set/subset used in constraints and objective terms.
ramp_cost _u	Technical	EUR/MW	Ramping capability/cost parameter.
ramp_down_cost_var	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
ramp_down_rate _u	Technical	p.u. per hour/year	Ramping capability/cost parameter.
ramp_up_cost_var	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
ramp_up_rate _u	Technical	p.u. per hour/year	Ramping capability/cost parameter.
rps_inv	Scenario flag	bool	Boolean switch that activates/deactivates a model block.
statCapInvCost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
statToPowert_cost	Objective auxiliary	EUR	Auxiliary expression used to compose the objective function.
technology_cap_costs_years _{u,y}	Technical	EUR/MW	Exogenous parameter or helper alias used in constraints/objective.
thermal_units	Set	-	Index set/subset used in constraints and objective terms.

Parameter / set (original model notation)	Category	Unit	Description
units	Technical	model units	Exogenous parameter or helper alias used in constraints/objective.
var_cost_{<i>u</i>}	Technical	EUR/MWh	Technical/economic parameter for generation technologies.
years	Set	-	Index set/subset used in constraints and objective terms.

Unit note: The model is hourly. Several power and energy quantities are numerically linked within each hour (MW average vs MWh per hour). Units above follow the most interpretable convention for each symbol.

5 Formulation by Equation Blocks and Their Role

Compact notation: $u \in \text{units}$, $t \in \text{periods}$, $y \in \text{years}$, $G_{u,t,y} = \text{generation}_{u,t,y}$, $I_{t,y} = \text{imports}_{t,y}$, $C_{t,y} = \text{ceep}_{t,y}$.

5.1 Objective Function

Description: This block aggregates all discounted annual cost components (operation, investment, flexibility, hydrogen, heat, and policy-related terms) into a single scalar objective Z .

$$\begin{aligned} \min Z = & \text{per_mw} + \text{import_cost} + \text{inv_cost} + \text{inv_cost_fossil} \\ & + \text{ramp_up_cost_var} + \text{ramp_down_cost_var} + \text{heat_pump_cap_cost} \\ & + \text{ResToHeat_cost} + \text{boiler_gen_cost} + \text{inv_cost_boiler} + \text{CO2_emissions_cost} \\ & + \text{evToPowert_cost} + \text{statToPowert_cost} + \text{statCapInvCost} \\ & + \text{h2_storage_cost} + \text{h2_CAPEX_cost} + \text{h2_OPEX_cost} \\ & + \text{inv_cost_FuelCell} + \text{per_mw_FC} + \text{industry_fuel_cost} \end{aligned} \quad (1)$$

Role: minimizes total discounted system cost across electricity, heat/cooling, hydrogen, transport flexibility, and investment decisions.

5.2 Policy and Scenario Activation Blocks

Description: These logical switches activate or deactivate optional model modules through equivalent algebraic constraints, enabling scenario analysis without changing the model core.

$$\text{NoResToHeatInv=True} \Rightarrow \text{heat_pump_cap} = 0 \quad (2)$$

$$\text{NoResToHeatInv=True} \Rightarrow \text{ResToHeat} = 0 \quad (3)$$

$$\text{NoResToHeatInv=True} \Rightarrow \text{heat_pump_out} = 0 \quad (4)$$

$$\text{rps_inv=True} \Rightarrow \text{RPS constraint is active} \quad (5)$$

$$\text{carbonLimit=True} \Rightarrow \text{annual CO2 cap is active} \quad (6)$$

$$\text{PrimaryReserve/SecondaryReserve=True} \Rightarrow \text{reserve modules are active} \quad (7)$$

Role: allows policy/scenario switching without changing the core model structure.

5.3 Primary Reserve (if active)

Description: The primary reserve module enforces hourly up/down reserve requirements and limits reserve provision using unit headroom/footroom and storage/flexibility feasibility conditions. **Role:** enforces short-term balancing capability.

5.4 Secondary Reserve (if active)

Description: This block mirrors the primary reserve logic with dedicated variables and constraints for secondary reserve products. **Role:** enforces additional balancing margin.

5.5 Industrial Heat Demand and Fuel Switching

Description: The equations below set a fuel-share upper bound per fuel and period, then close the industrial heat balance with endogenous hydrogen substitution.

$$\text{Fuel_ind}_{f,t,y} \leq \text{fuel_share_max}_{t,y,f} \cdot (\text{Industry_HT_demand}_{t,y} - \text{H2tInd}_{t,y}) \quad (8)$$

$$\sum_f \text{Fuel_ind}_{f,t,y} + \text{H2tInd}_{t,y} = \text{Industry_HT_demand}_{t,y} \quad (9)$$

Role: satisfies industrial thermal demand while allowing endogenous fuel mix with hydrogen substitution.

5.6 Hydrogen Production, Storage, and Demand Balance

Description: This block links electricity use to hydrogen production via efficiency, limits hydrogen conversion capacity, updates hydrogen storage SOC, and balances total hydrogen withdrawals with final uses.

$$\sum_{e \in \text{Elec_h2_units}} \frac{\text{generation_H2}_{e,t,y}}{\text{H2_eff}_e} = \text{power_to_h2}_{t,y} \quad (10)$$

$$\text{generation_H2}_{e,t,y} \leq \text{H2_init_cap}_e + \sum_{\tau \leq y} \text{h2_electrolysis_max_cap}_{e,\tau} \quad (11)$$

$$\text{h2_sto_soc}_{s,t,y} = \text{h2_sto_soc}_{s,t-1,y} - \text{h2_sto_out}_{s,t,y} + \sum_e \text{generation_H2}_{e,t,y} \quad (12)$$

with SOC bounds and total demand balance:

$$\sum_s \text{h2_sto_out}_{s,t,y} = \sum_q \text{H2_demand}_{q,t,y} + \sum_c \frac{\text{generation_FC}_{c,t,y}}{\text{FC_eff}_c} + \text{H2tInd}_{t,y} \quad (13)$$

Role: ensures physical consistency of the hydrogen subsystem.

5.7 EV and Stationary Storage Dynamics

Description: These equations track EV and stationary storage SOC trajectories over time with efficiency-adjusted charge/discharge flows, while enforcing SOC bounds, power limits, investment-based capacity limits, and end-of-horizon conditions.

5.7.1 EV Fleet Storage

Compact notation:

$$\begin{aligned} E_{t,y}^{ev} &\equiv \text{ev_sto_soc}_{t,y}, & P_{t,y}^{ev,in} &\equiv \text{ev_sto_in}_{t,y}, & P_{t,y}^{ev,out} &\equiv \text{ev_sto_out}_{t,y}, \\ \eta_y^{ev} &\equiv \text{ev_Grid_eff}, & \bar{E}_y^{ev} &\equiv \text{ev_sto_max}_y, & \alpha^{ev} &\equiv \text{ev_sto_min}, \\ \bar{P}_y^{ev} &\equiv \text{ev_Grid_P_max}_y, & a_{t,y}^{ev} &\equiv \text{ev_availability}_{t,y}, & L_{t,y}^{ev} \cdot D_y^{ev} &\equiv \text{ev_transp_load}_{t,y} \cdot \text{ev_Demand}_y. \end{aligned}$$

$$E_{1,y}^{ev} = -L_{1,y}^{ev} \cdot D_y^{ev} - P_{1,y}^{ev,out} + \eta^{ev} \cdot P_{1,y}^{ev,in} \quad (14)$$

$$E_{t,y}^{ev} = E_{t-1,y}^{ev} - L_{t,y}^{ev} \cdot D_y^{ev} - P_{t,y}^{ev,out} + \eta^{ev} \cdot P_{t,y}^{ev,in} \quad (15)$$

$$\alpha^{ev} \cdot \bar{E}_y^{ev} \leq E_{t,y}^{ev} \leq \bar{E}_y^{ev} \quad (16)$$

$$P_{t,y}^{ev,out} \leq \bar{P}_y^{ev} \cdot a_{t,y}^{ev} \quad (17)$$

$$P_{t,y}^{ev,in} \leq \bar{P}_y^{ev} \cdot a_{t,y}^{ev} \quad (18)$$

5.7.2 Stationary Battery Storage

Compact notation:

$$\begin{aligned} E_{b,t,y}^{st} &\equiv \text{stat_sto_soc}_{b,t,y}, & P_{b,t,y}^{st,in} &\equiv \text{stat_sto_in}_{b,t,y}, \\ P_{b,t,y}^{st,out} &\equiv \text{stat_sto_out}_{b,t,y}, & \eta_b^{st} &\equiv \text{Sta_sto_eff}_b, \\ \bar{E}_{b,y}^{st} &\equiv \text{Sta_sto_init_cap}_b \cdot \text{StaSto_Decom_ind}_y + \sum_{\tau \leq y} \text{stat_sto_cap_inv}_{b,\tau} \cdot \text{StaStoDecom}_{b,\tau,y}. \end{aligned}$$

$$E_{b,1,y}^{st} = -P_{b,1,y}^{st,out} + \eta_b^{st} \cdot P_{b,1,y}^{st,in} \quad (19)$$

$$E_{b,t,y}^{st} = E_{b,t-1,y}^{st} - P_{b,t,y}^{st,out} + \eta_b^{st} \cdot P_{b,t,y}^{st,in} \quad (20)$$

$$E_{b,t,y}^{st} \leq \bar{E}_{b,y}^{st} \quad (21)$$

$$\sum_y \text{stat_sto_cap_inv}_{b,y} \leq \text{Sta_sto_max_cap}_b \quad (22)$$

$$E_{b,T,y}^{st} \leq 0.2 \text{Sta_sto_max_cap}_b \quad (23)$$

Role: shifts load and supports balancing through electrochemical storage.

5.8 Generation Limits by Technology Type

Description: The block first defines effective available capacity from surviving existing assets plus surviving new investments, then applies technology-specific dispatch rules and annual investment ceilings.

$$\text{CapEff}_{u,y} = \text{max_load}_u \cdot \text{DecomOld}_{u,y} + \sum_{\tau \leq y} \text{cap_inv}_{u,\tau} \cdot \text{DecomNew}_{u,\tau,y} \quad (24)$$

$$\text{generation}_{u,t,y} \leq \text{cap_factor}_u \cdot \text{CapEff}_{u,y}, \quad u \in \text{fossil_units} \cup \text{hphs_units} \cup \text{hdam_units} \quad (25)$$

$$\text{generation}_{u,t,y} = \text{aval_factor_plant}_{u,t,y} \cdot \text{CapEff}_{u,y}, \quad u \in \text{nondisp_units} \cup \text{hror_units} \quad (26)$$

$$\text{cap_inv}_{u,y} \leq \text{max_inv_period}_u \quad (27)$$

Role: enforces physical and planning feasibility of generation.

5.9 Electricity Balance and CEPP Bound

Description: This block enforces hourly electricity adequacy by balancing injections (generation, imports, and storage discharge) with withdrawals (demand, charging, and sector-coupling electric loads). It also bounds annual `ceep` as a fraction of yearly electric-energy requirements. *Compact notation:*

$$\begin{aligned}
\mathcal{U}^{el} &\equiv \text{fossil_units} \cup \text{hphs_units} \cup \text{hdam_units} \cup \text{hror_units} \cup \text{nondisp_units}, \\
G_{t,y}^{tot} &\equiv \text{GenTot}_{t,y}, \quad G_{u,t,y} \equiv \text{generation}_{u,t,y}, \quad G_{c,t,y}^{FC} \equiv \text{generation_FC}_{c,t,y}, \\
I_{t,y} &\equiv \text{imports}_{t,y}, \quad D_{s,t,y} \equiv \text{demand}_{s,t,y}, \quad C_{t,y} \equiv \text{ceep}_{t,y}, \\
P_{t,y}^{ev,in} &\equiv \text{ev_sto_in}_{t,y}, \quad P_{t,y}^{ev,out} \equiv \text{ev_sto_out}_{t,y}, \quad \eta^{ev} \equiv \text{ev_Grid_eff}, \\
P_{b,t,y}^{st,in} &\equiv \text{stat_sto_in}_{b,t,y}, \quad P_{b,t,y}^{st,out} \equiv \text{stat_sto_out}_{b,t,y}, \quad \eta_b^{st} \equiv \text{Sta_sto_eff}_b, \\
H_{t,y} &\equiv \text{power_to_h2}_{t,y}, \quad R_{t,y} \equiv \text{ResToHeat}_{t,y}, \quad \bar{\alpha}^{ceep} \equiv \text{ceep_parameter}.
\end{aligned}$$

$$G_{t,y}^{tot} = \sum_{u \in \mathcal{U}^{el}} G_{u,t,y} + \sum_{c \in \text{FC_units}} G_{c,t,y}^{FC} \quad (28)$$

$$G_{t,y}^{tot} + I_{t,y} + \eta^{ev} \cdot P_{t,y}^{ev,out} + \sum_b \eta_b^{st} \cdot P_{b,t,y}^{st,out} = \sum_s D_{s,t,y} + \sum_b P_{b,t,y}^{st,in} + P_{t,y}^{ev,in} + H_{t,y} + R_{t,y} + C_{t,y} \quad (29)$$

$$\sum_t C_{t,y} \leq \bar{\alpha}^{ceep} \cdot \left(\sum_{s,t} D_{s,t,y} + \sum_{b,t} P_{b,t,y}^{st,in} + \sum_t R_{t,y} + \sum_t H_{t,y} + \sum_t P_{t,y}^{ev,in} \right) \quad (30)$$

Role: central electricity adequacy condition of the integrated system, plus an explicit annual quality/reliability bound through limited `ceep`.

5.10 Heat/Cooling Block (HP, Boilers, CHP)

Description: This block maps electricity-to-heat conversion via COP relations, enforces heat-pump and boiler operating limits, captures CHP heat-power coupling, updates thermal storage states, and closes hourly heat/cooling demand balances. *Compact notation:*

$$\begin{aligned}
Q_{h,m,t,y}^{dh} &\equiv \text{heat_pump_dh}_{h,m,t,y}, & Q_{h,m,t,y}^{cdh} &\equiv \text{cooling_pump_dh}_{h,m,t,y}, \\
Q_{h,t,y}^{ind} &\equiv \text{heat_pump_ind}_{h,t,y}, & Q_{h,t,y}^{cind} &\equiv \text{cooling_pump_ind}_{h,t,y}, \\
Q_{h,t,y}^{out} &\equiv \text{heat_pump_out}_{h,t,y}, & E_{h,t,y}^{hp} &\equiv \text{heat_pump_sto_soc}_{h,t,y}, \\
COP_{h,t,y} &\equiv \text{HP_COP_period}_{h,t,y}, & R_{t,y} &\equiv \text{ResToHeat}_{t,y}, \\
\bar{Q}_{h,m,y}^{dh} &\equiv \text{HPDHCapEff}_{h,m,y}, & \bar{Q}_{h,y}^{ind} &\equiv \text{HPINDCapEff}_{h,y}, \\
B_{b,m,t,y}^{dh} &\equiv \text{boiler_dh_generation}_{b,m,t,y}, & B_{b,t,y}^{ind} &\equiv \text{boiler_ind_generation}_{b,t,y}, \\
\bar{B}_{b,m,y}^{dh} &\equiv \text{BoilerDHCapEff}_{b,m,y}, & \bar{B}_{b,y}^{ind} &\equiv \text{BoilerIndCapEff}_{b,y}, \\
H_{m,t,y}^{in} &\equiv \text{heat_input}_{m,t,y}, & G_{m,t,y} &\equiv \text{generation}_{m,t,y}, \\
\kappa_m^{ht} &\equiv \text{CHPPowerToHeat}_m, & \kappa_m^{loss} &\equiv \text{CHPPowerLossFactor}_m, \\
\bar{G}_m &\equiv \text{max_load}_m \cdot \text{cap_factor}_m, & S_{m,t,y} &\equiv \text{heat_sto_level}_{m,t,y}, \\
Q_{m,t,y}^{gen} &\equiv \text{heat_gen}_{m,t,y}, & D_{m,t,y}^{heat} &\equiv \text{heat_demand}_{m,t,y}, \\
D_{t,y}^{heat,g} &\equiv \text{heat_demand}_{g^{heat},t,y}, & D_{t,y}^{cool,g} &\equiv \text{cooling_demand}_{g^{cool},t,y}, \\
g^{heat} &\equiv \text{general_demand}, & g^{cool} &\equiv \text{general_cooling}.
\end{aligned}$$

$$\sum_{h,m} \frac{Q_{h,m,t,y}^{dh} + Q_{h,m,t,y}^{cdh}}{COP_{h,t,y}} + \sum_h \frac{Q_{h,t,y}^{ind}}{COP_{h,t,y}} = R_{t,y} \quad (31)$$

$$Q_{h,m,t,y}^{dh} + Q_{h,m,t,y}^{cdh} \leq \bar{Q}_{h,m,y}^{dh} \quad (32)$$

$$Q_{h,t,y}^{ind} \leq \bar{Q}_{h,y}^{ind} \quad (33)$$

$$E_{h,t,y}^{hp} = E_{h,t-1,y}^{hp} + Q_{h,t,y}^{ind} - Q_{h,t,y}^{out} - Q_{h,t,y}^{cind} \quad (34)$$

$$B_{b,m,t,y}^{dh} \leq \bar{B}_{b,m,y}^{dh} \quad (35)$$

$$B_{b,t,y}^{ind} \leq \bar{B}_{b,y}^{ind} \quad (36)$$

$$H_{m,t,y}^{in} \cdot \kappa_m^{ht} \leq G_{m,t,y} \quad (37)$$

$$G_{m,t,y} \leq \bar{G}_m - H_{m,t,y}^{in} \cdot \kappa_m^{loss} \quad (38)$$

$$S_{m,t,y} = S_{m,t-1,y} + H_{m,t,y}^{in} + \sum_h Q_{h,m,t,y}^{dh} + \sum_b B_{b,m,t,y}^{dh} - Q_{m,t,y}^{gen} \quad (39)$$

$$Q_{m,t,y}^{gen} = D_{m,t,y}^{heat} \quad (40)$$

$$\sum_h Q_{h,t,y}^{out} + \sum_b B_{b,t,y}^{ind} = D_{t,y}^{heat,g} \quad (41)$$

$$\sum_h Q_{h,t,y}^{cind} = D_{t,y}^{cool,g} \quad (42)$$

Role: ensures thermally feasible operation and sector coupling consistency while preserving hourly heat/cooling service adequacy.

5.11 Hydro Reservoir and Pumped Hydro Storage

Description: This block models reservoir and pumped-hydro storage over time. It enforces storage bounds, water-balance recursions with inflow and efficiency, feasible withdrawals (generation plus spill), and end-of-horizon storage targets. *Compact notation:*

$$\begin{aligned} \mathcal{U}^{dam} &\equiv \text{hdam_units}, & \mathcal{U}^{phs} &\equiv \text{hphs_units}, & G_{u,t,y} &\equiv \text{generation}_{u,t,y}, \\ S_{u,t,y} &\equiv \text{storage_level}_{u,t,y}, & P_u &\equiv \text{max_load}_u, & O_{u,t,y} &\equiv \text{sto_out_flow}_{u,t,y}, \\ F_{u,t,y}^{dam} &\equiv \text{hdam_inflow}_{u,t,y}, & \eta_u^{dam} &\equiv \text{hdam_efficiency}_u, & \bar{S}_u^{dam} &\equiv \text{hdam_max_storage}_u, \\ F_{u,t,y}^{phs} &\equiv \text{hphs_inflow}_{u,t,y}, & \eta_u^{phs} &\equiv \text{hphs_efficiency}_u, & \bar{S}_u^{phs} &\equiv \text{hphs_max_storage}_u. \end{aligned}$$

$$0.5 \bar{S}_u^{dam} \leq S_{u,t,y} \leq \bar{S}_u^{dam}, \quad u \in \mathcal{U}^{dam} \quad (43)$$

$$S_{u,1,y} = 0.5 \bar{S}_u^{dam} + F_{u,1,y}^{dam} \cdot P_u \cdot \eta_u^{dam} - \frac{G_{u,1,y}}{\eta_u^{dam}} - O_{u,1,y} \quad (44)$$

$$S_{u,t,y} = S_{u,t-1,y} + F_{u,t,y}^{dam} \cdot P_u \cdot \eta_u^{dam} - \frac{G_{u,t,y}}{\eta_u^{dam}} - O_{u,t,y} \quad (45)$$

$$\frac{G_{u,t,y}}{\eta_u^{dam}} + O_{u,t,y} \leq S_{u,t,y} + F_{u,t,y}^{dam} \cdot P_u \cdot \eta_u^{dam} \quad (46)$$

$$S_{u,T,y} \geq 0.5 \bar{S}_u^{dam} \quad (47)$$

$$0.5 \bar{S}_u^{phs} \leq S_{u,t,y} \leq \bar{S}_u^{phs}, \quad u \in \mathcal{U}^{phs} \quad (48)$$

$$S_{u,t,y} = S_{u,t-1,y} + F_{u,t,y}^{phs} \cdot P_u \cdot \eta_u^{phs} - \frac{G_{u,t,y}}{\eta_u^{phs}} - O_{u,t,y} \quad (49)$$

$$\frac{G_{u,t,y}}{\eta_u^{phs}} + O_{u,t,y} \leq S_{u,t,y} + F_{u,t,y}^{phs} \cdot P_u \cdot \eta_u^{phs} \quad (50)$$

$$S_{u,T,y} \geq 0.5 \bar{S}_u^{phs} \quad (51)$$

Role: preserves physically feasible hydro energy trajectories and prevents end-horizon reservoir depletion.

5.12 Ramping, Imports, RPS, and CO2 Policy

Description: This block combines short-term operational constraints (ramping and import smoothing) with strategic policy constraints (renewable portfolio share and annual CO2 cap). *Compact notation:*

$$\begin{aligned} \mathcal{U}^{disp} &\equiv \text{disp_units}, \quad \mathcal{U}^{rps} \equiv \text{nondisp_units} \cup \text{biomass_units} \cup \text{hydro_units}, \\ \mathcal{U}^{el} &\equiv \text{fossil_units} \cup \text{hphs_units} \cup \text{hdam_units} \cup \text{hror_units} \cup \text{nondisp_units}, \\ G_{u,t,y} &\equiv \text{generation}_{u,t,y}, \quad I_{t,y} \equiv \text{imports}_{t,y}, \quad \bar{I}_t \equiv \text{import_ntc}_t, \\ r_u^{up} &\equiv \text{ramp_up_rate}_u, \quad r_u^{down} \equiv \text{ramp_down_rate}_u, \quad \rho_y \equiv \text{rps}_y, \\ E_y^{el} &\equiv \text{CO2_total}_y, \quad E_y^{ht} \equiv \text{CO2_total_heat}_y, \quad E_y^{ind} \equiv \text{CO2_total_industry}_y, \\ \bar{E}_y &\equiv \text{CO2_limit}_y, \quad \phi_u \equiv \text{CO2_factor}_u, \quad \phi_b^{th} \equiv \text{Thermal_CO2}_b, \\ \phi_f^f &\equiv \text{Fuel_CO2}_f, \quad B_{b,t,y}^{ind} \equiv \text{boiler_ind_generation}_{b,t,y}, \\ B_{b,m,t,y}^{dh} &\equiv \text{boiler_dh_generation}_{b,m,t,y}, \quad F_{f,t,y} \equiv \text{Fuel_ind}_{f,t,y}. \end{aligned}$$

$$G_{u,t,y} \leq G_{u,t-1,y} + r_u^{up} \cdot \text{CapEff}_{u,y}, \quad u \in \mathcal{U}^{disp} \quad (52)$$

$$G_{u,t,y} \geq G_{u,t-1,y} - r_u^{down} \cdot \text{CapEff}_{u,y}, \quad u \in \mathcal{U}^{disp} \quad (53)$$

$$I_{t,y} \leq \bar{I}_t \quad (54)$$

$$I_{t,y} \leq 1.2 I_{t-1,y} \quad (55)$$

$$I_{t,y} \geq 0.8 I_{t-1,y} \quad (56)$$

$$\sum_{u \in \mathcal{U}^{rps}} \sum_t G_{u,t,y} \geq \rho_y \left(\sum_{u \in \mathcal{U}^{el}} \sum_t G_{u,t,y} + \sum_t I_{t,y} \right) \quad (57)$$

$$E_y^{el} = \sum_{u \in \text{fossil_units}, t} \phi_u \cdot G_{u,t,y} \quad (58)$$

$$E_y^{ht} = \sum_{b,t} \phi_b^{th} \cdot B_{b,t,y}^{ind} + \sum_{b,m,t} \phi_b^{th} \cdot B_{b,m,t,y}^{dh} \quad (59)$$

$$E_y^{ind} = \sum_{f,t} \phi_f^f \cdot F_{f,t,y} \quad (60)$$

$$E_y^{el} + E_y^{ht} + E_y^{ind} \leq \bar{E}_y \quad (61)$$

Role: preserves feasible system operation hour-by-hour while simultaneously enforcing portfolio and emissions policy trajectories.

5.13 Ramping Cost Linearization

Description: Auxiliary ramp-cost variables are lower-bounded by signed generation changes, yielding a linear representation of up/down ramping penalties in the objective.

$$\text{rampup_cost}_{u,t,y} \geq \text{ramp_cost}_u (G_{u,t,y} - G_{u,t-1,y}) \quad (62)$$

$$\text{rampdown_cost}_{u,t,y} \geq \text{ramp_cost}_u (-G_{u,t,y} + G_{u,t-1,y}) \quad (63)$$

Role: penalizes fast dispatch changes while preserving linearity.

6 Implementation Notes (Current Code Traceability)

- `Power_to_h2` is created with an extra loop over `flex_unit`; mathematically it is equivalent to one equation per (t, y) but instantiated repeatedly.
- `primary_hourly_reserve_down_units_minimum2` and `secondary_hourly_reserve_down_units_minimum2` use an up-reserve variable on the left-hand side.
- `C02_cost_industry` is computed but not included in the final objective sum.
- With `Imp_price_inc=0.05`, the expression $(1 + \text{Imp_price_inc}/100)^n$ corresponds to 0.05% annual growth.

A Appendix A: Reserve Module Data Dictionary

This appendix consolidates the data dictionary entries used specifically in Sections 4.3 and 4.4.

A.1 A.1 Decision Variables for Reserve Modules

Variable (original model notation)	Type	Unit	Description
primary_hourly_reserve_up_units _{u,t,y}	Reserve	MW _e	Upward primary reserve from dispatchable units.
primary_hourly_reserve_down_units _{u,t,y}	Reserve	MW _e	Downward primary reserve from dispatchable units.
primary_hourly_stat_stoOUT_reserve_up_units _{b,t,y}	Reserve	MW _e	Upward primary reserve via stationary storage discharge margin.
primary_hourly_stat_stoOUT_reserve_down_units _{b,t,y}	Reserve	MW _e	Downward primary reserve via stationary storage discharge margin.
primary_hourly_stat_stoIN_reserve_up_units _{b,t,y}	Reserve	MW _e	Upward primary reserve via stationary storage charging margin.
primary_hourly_stat_stoIN_reserve_down_units _{b,t,y}	Reserve	MW _e	Downward primary reserve via stationary storage charging margin.
primary_hourly_FC_reserve_up _{c,t,y}	Reserve	MW _e	Upward primary reserve from fuel cells.
primary_hourly_FC_reserve_down _{c,t,y}	Reserve	MW _e	Downward primary reserve from fuel cells.
primary_hourly_ELY_reserve_down _{e,t,y}	Reserve	MW _e	Downward primary reserve from electrolyzers.
primary_hourly_ELY_reserve_up _{e,t,y}	Reserve	MW _e	Upward primary reserve from electrolyzers.
primary_hourly_HPind_reserve_down _{h,t,y}	Reserve	MW _e	Downward primary reserve from industrial heat pumps.
primary_hourly_HPind_reserve_up _{h,t,y}	Reserve	MW _e	Upward primary reserve from industrial heat pumps.
secondary_hourly_reserve_up_units _{u,t,y}	Reserve	MW _e	Upward secondary reserve from dispatchable units.
secondary_hourly_reserve_down_units _{u,t,y}	Reserve	MW _e	Downward secondary reserve from dispatchable units.
secondary_hourly_stat_stoOUT_reserve_up_units _{b,t,y}	Reserve	MW _e	Upward secondary reserve via stationary storage discharge margin.
secondary_hourly_stat_stoOUT_reserve_down_units _{b,t,y}	Reserve	MW _e	Downward secondary reserve via stationary storage discharge margin.
secondary_hourly_stat_stoIN_reserve_up_units _{b,t,y}	Reserve	MW _e	Upward secondary reserve via stationary storage charging margin.
secondary_hourly_stat_stoIN_reserve_down_units _{b,t,y}	Reserve	MW _e	Downward secondary reserve via stationary storage charging margin.
secondary_hourly_FC_reserve_up _{c,t,y}	Reserve	MW _e	Upward secondary reserve from fuel cells.
secondary_hourly_FC_reserve_down _{c,t,y}	Reserve	MW _e	Downward secondary reserve from fuel cells.
secondary_hourly_ELY_reserve_down _{e,t,y}	Reserve	MW _e	Downward secondary reserve from electrolyzers.
secondary_hourly_ELY_reserve_up _{e,t,y}	Reserve	MW _e	Upward secondary reserve from electrolyzers.
secondary_hourly_HPind_reserve_down _{h,t,y}	Reserve	MW _e	Downward secondary reserve from industrial heat pumps.
secondary_hourly_HPind_reserve_up _{h,t,y}	Reserve	MW _e	Upward secondary reserve from industrial heat pumps.

A.2 A.2 Parameters and Sets for Reserve Modules

Parameter / set (original model notation)	Category	Unit	Description
PrimaryReserve	Scenario flag	bool	Activates the primary reserve constraint block.
SecondaryReserve	Scenario flag	bool	Activates the secondary reserve constraint block.
res_inv	Scenario flag	bool	Activates reserve-investment related module switches.
reserve_factor _{<i>u,t,y</i>}	Technical	p.u.	Reserve-down scaling factor applied at the same hourly granularity as reserve variables (often time-invariant in data).
primary_reserve_units	Set	-	Index set for <i>u</i> in primary reserve variables primary_hourly_reserve_up_units _{<i>u,t,y</i>} and primary_hourly_reserve_down_units _{<i>u,t,y</i>} .
primary_reserve_StatStorage_units	Set	-	Index set for <i>b</i> in stationary-storage primary reserve variables.
primary_reserve_FC_units	Set	-	Index set for <i>c</i> in fuel-cell primary reserve variables.
primary_reserve_ELY_units	Set	-	Index set for <i>e</i> in electrolyzer primary reserve variables.
primary_reserve_HP_units	Set	-	Index set for <i>h</i> in industrial heat-pump primary reserve variables.
secondary_reserve_units	Set	-	Index set for <i>u</i> in secondary reserve variables secondary_hourly_reserve_up_units _{<i>u,t,y</i>} and secondary_hourly_reserve_down_units _{<i>u,t,y</i>} .
secondary_reserve_StatStorage_units	Set	-	Index set for <i>b</i> in stationary-storage secondary reserve variables.
secondary_reserve_FC_units	Set	-	Index set for <i>c</i> in fuel-cell secondary reserve variables.
secondary_reserve_ELY_units	Set	-	Index set for <i>e</i> in electrolyzer secondary reserve variables.
secondary_reserve_HP_units	Set	-	Index set for <i>h</i> in industrial heat-pump secondary reserve variables.